DE–2983

Sub. Code
101

DISTANCE EDUCATION

M.C.A. DEGREE EXAMINATION, DECEMBER 2017.

First Semester

DIGITAL COMPUTER ORGANIZATION

(2010 Academic Year onwards)

Time : Three hours Maximum : 100 marks

PART A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. Convert the following :

   (a) $(1A5D)_{16}$ into Decimal.

   (b) $(10767)_{10}$ into Hexadecimal.

2. State and prove Demorgan's theorem.

3. What is demultiplexer? Discuss the functions of a demultiplexer.

4. What is an XOR gate? Write its T.T and realize it using AND, OR and NOT gates.

5. Draw the circuit diagram of a shift register and explain its working in detail.

6. Explain in short about counters.

7. Distinguish between Static and Dynamic RAM.

8. What are the different steps used for executing an instruction? Explain.

PART B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Using Five variable K-map Simplify $f = (A, B, C, D, E) =$

$\sum(0, 2, 3, 4, 6, 7, 8, 12, 14, 15, 16, 18, 19, 20, 22, 23, 24, 28)$

10. Explain the principles of Universal shift register with a neat schematic diagram.

11. Explain Up Down parallel counter with necessary diagram.

12. What is the difference between the synchronous and asynchronous counters? Explain with neat diagram about asynchronous counter.

13. Draw the schematic diagram of a master-slave j-k flip-flop. Discuss its working principle. What are its advantages over other type of flip-flop?

14. What are interrupts? Why are they used? Explain what happens when a computer recognizes an interrupt?

15. (a) Write short notes on I/O interfaces.

(b) Explain DMA in detail.

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DE–2984

Sub. Code
102

DISTANCE EDUCATION

M.C.A. DEGREE EXAMINATION, DECEMBER 2017.

First Semester

C AND DATA STRUCTURES

(2010 Academic Year Onwards)

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. Explain the Nested if-else statement by giving an example.

2. Write short notes on Constant, Variables and their types.

3. Explain the procedure for initializing 2-D arrays.

4. Write a note on Recursion.

5. Explain the use of Structures in C by giving an example.

6. Define the terms Pointers.

7. Discuss about the applications of Queue.

8. Write a note on Doubly linked list.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Explain the input/output statements available in C.

10. Write a C program to arrange 'n' numbers in ascending order.

11. Explain how to create and processing a data file?

12. Explain the operations that can be performed on Stacks.

13. Explain insertion and deletion in doubly linked lists by giving the algorithms.

14. Write a program to store student details in a file.

15. Write a C program to generate a mark sheet using file handling.

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DE–2985

Sub. Code
103

DISTANCE EDUCATION

MASTER OF COMPUTER APPLICATION DEGREE EXAMINATION, DECEMBER 2017.

First Semester

RELATIONAL DATABASE MANAGEMENT SYSTEMS

(2010 Academic year onwards)

Time : Three hours Maximum : 100 marks

PART A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. What are the different user interfaces? Discuss.

2. Specify the manipulative capabilities of the DBMS.

3. Briefly discuss the mapping operation of relational model.

4. Describe the navigation and efficiency of relational model.

5. What are the relational database's properties? Explain.

6. Write about the client/server systems.

7. Mention the data access SQL commands.

8. What are the embedded SQL statements available? Describe.

PART B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Explain the database management system in detail with neat diagram.

10. Discuss the data control and data definition facilities.

11. Give an overview of oracle database architecture with neat diagram.

12. Discuss the basic queries and sub queries with suitable examples.

13. Write the queries to perform the following:

    (a) Create table (3)

    (b) Insert row into a table (3)

    (c) Copying another table (3)

    (d) Rename table (3)

    (e) Delete table. (3)

14. Explain in detail the relational model based DBMS.

15. What are the languages supported by oracle precompiler? Explain in detail.

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DE–2986

Sub. Code
104

DISTANCE EDUCATION

M.C.A. DEGREE EXAMINATION, DECEMBER 2017.

First Semester

OPERATING SYSTEMS

(2010 Academic Year onwards)

Time : Three hours Maximum : 100 marks

PART A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. Define operating system. Explain the different types of operating system.

2. What is Semaphore? Explain the role of semaphore in implementing mutual exclusion.

3. Write short notes on the following :

   (a) Monitors

   (b) Message passing.

4. What do you mean by interrupts? How the interrupts are handled by an operating System?

5. Explain in detail about device drivers.

6. Describe the multiprogramming with variable partition.

7. Discuss the LRU page replacement algorithm with an example.

8. What is file system? Explain briefly.

PART B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. (a) Discuss the services provided by the operating system.

   (b) What are system calls? Explain any five system calls.

10. Write short notes on the following :

    (a) Race conditions

    (b) Critical sections

    (c) Mutual exclusion

11. Explain the priority scheduling and shortest job first scheduling with illustration.

12. Discuss the functions of device-independent I/O software in detail.

13. What do you mean by deadlock? Discuss in detail about deadlock avoidance algorithm with neat diagrams.

14. Discuss in detail about paging and page tables.

15. Explain in detail about the protection mechanisms that are used in operating systems to protect files.

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DE–9844

Sub. Code
105

DISTANCE EDUCATION

M.C.A. DEGREE EXAMINATION, DECEMBER 2017.

First Semester

DATA STRUCTURE AND USING C LAB

(2010 Academic Year onwards)

Time : Three hours Maximum : 100 marks

Examiner has to select ONE question and give it to the student by a lot system.

Each subdivision carries 50 marks.

1. (a) Write a program in C using function to subtract matrix A from matrix B.

   (b) Write a program to implement the following operations on a doubly linked list.

      (i) Insert a new node in a list.

      (ii) Count the number of nodes in the list.

---------- Cut here ----------

2. (a) Write a program to find the roots of a quadratic equation.

   (b) Write a Program to implement the PUSH and POP operations on Stack.

3. (a) Write a program to identify whether a given word is PALINDROME or not.

(b) Write a program to implement the following operations on a singly linked list.

(i) Insert new node at the beginning of a list

(ii) Delete the Middle node.

(iii) Count the number of nodes in the list.

---------- Cut here ----------

4. (a) Write a program to convert an Infix notation to Postflx using Stack.

(b) Write a program in C to sort N elements using Selection sort.

---------- Cut here ----------

5. (a) Implement the insert and delete operations on Queue using arrays.

(b) Write a program to evaluate the series, $1 + x + x^2 + x^3 + x^6 + x^7 + x^{10} + ...$

6. (a) Calculate the Electricity Bill for domestic user as per the following slap :

Units	Tariff
<100	Rs. 5/- (Fixed rate)
≥ 100 and <200	Rs. 1.20/-
≥ 200 and <300	Rs. 2.75/-
≥ 300 and <500	Rs. 3.20/-
≥ 500	Rs. 6/-

(b) Write a program to perform the following operations:

(i) Create a file.

(ii) Get data from user through Keyboard.

(iii) Stores the data into the file.

(iv) Display the stored information on the monitor.

---------- Cut here ----------

7. (a) Write a program which displays the numbers from 1 to 1000 that are divisible by 3.

(b) Write a menu driven program to find a particular element in a set of elements based on Linear and Binary search techniques.

---------- Cut here ----------

8. (a) Write a program to sort the percentage of marks awarded to your class mate with Course name, year, subject name and code.

(b) Write a program in C to sort N elements using Insertion sort.

9. (a) Write a program to generate the Fibonacci series up to N numbers.

(b) Develop a program to add the following polynomials using linked list.

$$X^3 + 4X^2 + 2X + C = 0$$

$$5X^3 + X^2 + 6X + D = 0.$$

DE–9845

Sub. Code
106

DISTANCE EDUCATION

MASTER OF COMPUTER APPLICATION DEGREE EXAMINATION, DECEMBER 2017.

First Semester

RELATIONAL DATABASE MANAGEMENT SYSTEMS LAB

(2010 Academic Year Onwards)

Time : Three hours Maximum : 100 marks

Examiner has to select ONE question and give it to the students by the lot system.

1. A Newspaper vendor buys newspapers at wholesale from a distributor for Rs. 4 and sell them at retail for Rs. 4.50. At the end of the day the unsold papers are returned to the distributor for Rs. 1 rebate per paper. Write a program using oracle to prepare a report for the newspaper vendor in the following format with 9 weeks data:

Week	Bought	Sold	Return	Profit	Loss
1	100	50	XX	XXX	XX
2	100	50	XX	XXX	XX
.	.....	.....	.....	.....	.....
.	.....	.....	.....	.....	.....
Total					

2. An airline reservation database contains the reservation table and personal table.

Reservation table contains following fields :

Flight_no, class, passenger_name, seat_number.

Personal table contains following fields :

Passenger_name, sex, age, marital_status, nationality.

Write a program in oracle to prepare the following reports:

(a) List the passenger names with seat number according to class and flight number.

(b) The total number of married female passengers on a particular flight.

(c) List out the male passengers in the age between 18 to 25 for all flights.

---------- Cut here ----------

3. A hospital maintains blood donor record information in a file. The file contains the following fields :

Donor_number, donor_name, donor_age, donor_address, pincode, place_of_birth, sex, blood_group (*A, B, AB, O*).

Write SQL statements to print Donor's name, number, and address for the following categories.

(a) Blood donors having 'AB' blood group.

(b) Donors in the age between 16 to 25.

(c) Female donors with 'A' blood group in the age between 18 to 25.

4. Create a table MARKS contains:

Roll no, name, mark 1, mark 2, mark 3

Using the MARKS table do the following:

(a) Search a Roll no and display total marks and grade.

(b) For a pass subject minimum is 40. For the failed students grade is "F". For the passed student grade is set as

'A' if total > 240

'B' if total between 180 and 240

'C' if total between 150 and 180

'D' otherwise exception must be raised if Roll no could not find.

---------- Cut here ----------

5. Create a table to represent a theatre ticket reservation consisting of the following fields.

Customer_name, sex, date, seat_number, show_time, class.

(a) Create the above table by properly specifying primary keys and foreign keys.

(b) Find the customers who have more than 3 tickets.

(c) Display all the female customers in a show.

(d) Display the shows which are houseful in a day.

6. A daily sales file contains records with the following fields.

   Department_number, date, item_description, quantity, cost_of_each_item, sale_price_for_each_item.

   Write a program using Oracle to list all the input data and compute the total amount of the sales and profit for each sale. Accumulate and print total sales and profit. Show the output report in the following format:

   Department number:
   Item Description:
   Sales Price:
   Quantity:
   Cost:
   Profit:

---------- Cut here ----------

7. Write a PL/SQL program to prepare a mark statement of a student using Student table and Mark table. Assume your own data.

---------- Cut here ----------

8. Write a program in SQL for hospital billing system with the following fields:

   patient_no, patient_name, age, attending_doctor, patient_type (*In/Out*), consulting_charge, blood_test_charges, Xray_charges, other_test_charges, and total_fee.

   Prepare reports with the following conditions:

   (a) The patients, undergone blood test.
   (b) The patients, who taken X-ray.
   (c) The patients, who belong to In-patient category.

   The common fields to be included in the above mentioned reports:

   patient_no, patient_name, age, charges, total_fee.

DE–2987

Sub. Code
201

DISTANCE EDUCATION

M.C.A. DEGREE EXAMINATION, DECEMBER 2017.

Second Semester

OBJECT ORIENTED PROGRAMMING AND C++

(2010 Academic Year onwards)

Time : Three hours Maximum : 100 marks

PART A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. What are the principles of object oriented programming?

2. Explain the various data types available in C++ with suitable example.

3. Explain the various looping statements available in C++ with example.

4. Write a C++ program to find a factorial of a given n number.

5. Explain the 'this' pointer with an example.

6. Explain the multilevel inheritance with example.

7. Discuss in detail about pure virtual function with an example.

8. Describe the command line arguments.

PART B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Explain the structure of a C++ program with an example.

10. Explain the constructor and destructor with an example.

11. Write a C++ program to sort numbers in Ascending order using pointer concept.

12. Write a C++ program for binary operator overloading.

13. Explain the Friend Function with an example program.

14. Describe the abstract class with example.

15. Write a note on function overloading and data hiding.

DE–2988

Sub. Code
202

DISTANCE EDUCATION

M.C.A. DEGREE EXAMINATION, DECEMBER 2017.

Second Semester

UNIX AND SHELL PROGRAMMING

(2010 Academic Year Onwards)

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. Discuss the components of UNIX and kernel-shell relationship.

2. Explain Sed editor and its basic operations.

3. Write short notes on Directories and inodes.

4. Write short notes on standard input and output in Unix.

5. Discuss about shell and execution of shell script with an example.

6. What do you mean by filters? Give examples of filters.

7. Explain MS Macro Package.

8. Explain the following with suitable
   (a) chmod
   (b) unmask.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. (a) What are the basic functions of shell?

(b) Explain the different types of shells used in UNIX OS.

10. (a) Why is UNIX more portable than any other operating system?

(b) Discuss its multi-user and multi-tasking features.

11. (a) Discuss different operational modes of vi editor.

(b) Generate line numbers for the file being edited currently, mark a block of lines and perform cut and paste.

12. (a) What is system call? Discuss about any four system calls.

(b) Discuss about file permission and its types.

13. What are the basic functions of a shell? Explain the different types of shells used in UNIX OS?

14. (a) Explain conditional and looping structures in shell programming.

(b) Write a program to reverse the given number which consists of n digits.

15. Write short notes on :

(a) Pattern matching

(b) Error handling with suitable examples.

DE–2989

Sub. Code
203

DISTANCE EDUCATION

M.C.A. DEGREE EXAMINATION, DECEMBER 2017.

Second Semester

COMMUNICATION SKILLS

(2010 Academic Year onwards)

Time : Three hours Maximum : 100 marks

PART A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. Explain the process of Communication.

2. How could you improve your art of writing in English?

3. Write a note on the art of listening.

4. Define Mock Interview.

5. Explain the process of group discussion.

6. Bring out the objectives of meeting

7. What is interpersonal behaviour?

8. Define semantics.

PART B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Explain the commandments of communication.

10. Discuss linear reading.

11. How could you improve your art of conversation?

12. What are the tools used for personality identification in body language? Explain.

13. Describe the importance of Mock Group Discussion.

14. How could you improve your communication skills for meetings?

15. Explain communication within management.

DE–2990

Sub. Code
204

DISTANCE EDUCATION

M.C.A. DEGREE EXAMINATION, DECEMBER 2017.

Second Semester

COMPUTER GRAPHICS

(2010 Academic Year onwards)

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. Write a note on the following :
   (a) Joy Stick
   (b) Light Pen
   (c) Plotter
   (d) LED monitor.

2. Explain the Midpoint circle generation algorithm.

3. Write a note on Line segmentation.

4. Discuss the different types of view ports.

5. What is meant by Clipping? Discuss with an example.

6. Explain the Two dimensional transformations.

7. Explain the different types of projections.

8. Illustrate the User's model.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Write an algorithm to draw an ellipse.

10. Discuss the Region filling technique.

11. Describe the three-dimensional transformations with matrix notation.

12. Explain the Covex polygon clipping algorithm

13. Write a detailed note on Mirror reflection.

14. Explain the Cohen Sutherland algorithm for line clipping.

15. What is command language? Discuss the various styles of command language.

DE–9846

Sub. Code
205

DISTANCE EDUCATION

M.C.A. DEGREE EXAMINATION, DECEMBER 2017.

Second Semester

C++ AND GRAPHICS LAB

(2010 Academic year onwards)

Time : Three hours

Maximum : 100 marks

Record Note Book : 5
Algorithm/Flow chart : 15
Program : 35
Execution : 35
Result : 10
Total : 100

Examiner has to choose ONE question for each candidate by lot system.

Each sub-division carries 50 marks.

1. (a) Write a C++ program to draw a straight line using Bresenham's line drawing algorithm.

   (b) Write a C++ program to draw an ellipse.

---------- Cut here ----------

2. (a) Create a base class shape, which contains the common element of circle from which you derive a class called ball. Include two member functions set( ) to set coordinates and draw( ) which draws the circle. Write a main( ) program to exercise this class using graphic-mode graphic functions.

   (b) Write a C++ program to sketch a triangle.

3. (a) Develop a program using functions to perform the following :

   (i) Draw a rectangle.

   (ii) Rotate it through 90 degree

   (iii) Translate to another position.

   (b) Fill the rectangle drawn in (a) with red blue colour.

---------- Cut here ----------

4. (a) Write a C++ program to draw a straight line using Bresnham's line drawing algorithm.

   (b) Write a C++ program to sketch a triangle.

---------- Cut here ----------

5. Develop a program to do the following :

   (a) Draw a triangle ABC, translate it to a distance and then rotate it through an angle clock-wise.

   (b) Draw a polygon.

---------- Cut here ----------

6. (a) Write a program in C++ to draw Concentric circle with three objects.

   (b) Draw a ball, and make animation effect on it.

7. (a) Write a C++ program to perform graphical transformation (Scaling. Rotation, Translation) of a two dimensional object square.

(b) Draw a ball, and make animation effect on it.

---------- Cut here ----------

8. (a) Write a C++ program to draw triangle using SHADING technique.

(b) Write a program in C++ to clip a part of a line.

---------- Cut here ----------

9. (a) Draw a Circle using Bresenham's algorithm.

(b) Make animation effect on the Circle drawn in (a).

---------- Cut here ----------

10. (a) Develop a program using functions to perform the following :

(i) Draw a rectangle

(ii) Rotate it through 90 degree

(iii) Translate to another position.

(b) Fill the rectangle drawn in (a) with red colour.

11. (a) Draw a ball, and make animation effect on it.

(b) Draw a triangle ABC, translate it to a distance and then rotate it through an angle Clock-wise.

---------- Cut here ----------

12. (a) Develop a program using functions to perform the following :

(i) Draw a Polygon.

(ii) Rotate it through 90 degree

(iii) Translate to another position.

(b) Fill the Polygon drawn in (a) with red colour.

---------- Cut here ----------

13. (a) Create a base class shape, which contains the common element of circle from which you derive a class called ball. Include two member functions set( ) to set coordinates and draw( ) which draws the circle. Write a main( ) program to exercise this class using graphic-mode graphic functions.

(b) Write a program in C++ to draw a polygon.

---------- Cut here ----------

14. (a) Draw a Circle using Bresenham's algorithm. Divide the circle into four quadrant, and fill with different colours.

(b) Make animation effect on the Circle drawn in (a).

---------- Cut here ----------

15. (a) Draw a ball, and make animation effect on it.

(b) Draw a triangle ABC, translate it to a distance and then rotate it through an angle Clock-wise.

DE–9847

Sub. Code
206

DISTANCE EDUCATION

M.C.A. DEGREE EXAMINATION, DECEMBER 2017.

Second Semester

UNIX AND SHELL PROGRAMMING LAB

(2010 Academic Year Onwards)

Time : Three hours Maximum : 100 marks

Examiner has to select ONE question and give it to the student by the lot system.

1. (a) Write a shell script that accepts a file name starting and ending line numbers as arguments and displays all the lines between the given line numbers.

   (b) Write a shell script that deletes all lines containing a specified word in one or more files supplied as-arguments to it.

---------- Cut here ----------

2. (a) Write a shell script that computes the gross salary of a employee according to the following rules:

   (i) If basic salary is < 1,500 then HRA = 10% of the basic and DA = 90% of the basic.

   (ii) If basic salary is >= 1,500 then HRA = Rs. 500 and DA = 98% of the basic.

   The basic salary is entered interactively through the key board.

   (b) Write a shell program to reverse the given string.

3. (a) Write a shell script that accepts two integers as its arguments and computes the value of first number raised to the power of the second number.

   (b) Write an interactive file-handling shell program, which offers the choice of copying, removing. renaming, linking files.

---------- Cut here ----------

4. (a) A school maintains students' details. Write a shell program to list the following:

   (i) Students listed as male and female.

   (ii) Students marks above 70 and marks below 40.

   (iii) Students' name, address, phone numbers.

   (b) Write a program for four-function calculator by using yacc command

---------- Cut here ----------

5. (a) Write a shell script which receives two file names as arguments. It should check whether the two file contents are same or not. If they are same then second file should be deleted.

   (b) Write a shell script that takes a command – line argument and reports on whether it is directory, a file or something else.

6. (a) Write and execute the UNIX Commands for performing the following:

   (i) To create a directory.

   (ii) To compare two files and show the differences.

   (iii) To pipe the output of "who" command to the "sort" command.

   (iv) To change the ownership of a file

   (v) To save the output in a file and display the same on the VDU.

   (b) Write a shell script to display user name, current directory, current date and time.

---------- Cut here ----------

7. (a) Write a menu driven shell program for the following:

   (i) To check if a file is writ able.

   (ii) Exit from Unix.

   (b) Write a shell program to accept a number from 0 to 9 and display the same number in words.

---------- Cut here ----------

8. (a) Write a script which reads a number in units of seconds and converts it to the units hours:minutes:seconds and prints the result to standard output.

   (b) Write a shell program to accept a string and check whether it is palindrome or not.

9. (a) Write a shell program to arrange the names in alphabetical order with swapping done in a separate function.

(b) Write a shell program to generate the Fibonacci series.

---------- Cut here ----------

10. (a) Write a shell program for the following:

(i) List the files in date order.

(ii) Sort the files in descending order.

(b) Write a shell program to check whether given number is even or odd using positional parameters.

DE–2991

Sub. Code
301

DISTANCE EDUCATION

M.C.A./M.C.A. (Lateral) DEGREE EXAMINATION, DECEMBER 2017.

Third Semester

DISCRETE MATHEMATICS

(2010 Academic Year Onwards)

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. Show the equivalence : $P \to (Q \to P) \Leftrightarrow \rceil P \to (P \to Q)$.

2. Show that $S \vee R$ is tautologically implied by

   $(P \vee Q) \wedge (P \to R) \wedge (Q \to S)$.

3. Define the following with an example

   (a) Universal set

   (b) Proper subset

   (c) Equal set.

4. If $X = \{1,2,.....7\}$ and $R = \{< x, y > | \, x - y$ is divisible by 3$\}$ show that R is an equivalence relation.

5. Write a note on hashing function with an example.

6. If $< M, * >$ is any commutative monoid, then show that the set of idempotent elements of M forms a submonoid.

7. Show that every cyclic group of order n is isomorphic to the group $< z_n, +_n >$.

8. Define the following with an example.

   (a) Graph

   (b) Connected graph

   (c) Sub graph

   (d) Directed tree.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. (a) Obtain the principal conjunctive normal form of the formula S given by $(\neg P \to R) \wedge (Q \rightleftarrows P)$.

   (b) Obtain the principal disjunctive normal form of the formula $(P \to (Q \wedge R) \wedge (\neg P \to (\neg Q \wedge \neg R))$.

10. (a) Show that $(x)(P(x) \vee Q(x)) \Rightarrow (x)P(x) \vee (\exists(x))Q(x)$.

    (b) Show the following by constructing derivations.

    $(x)(P(x) \to (Q(y) \wedge R(x)))$,
    $(\exists(x)\ P(x) \Rightarrow Q(y) \wedge (\exists x)(P(x) \wedge R(x))$.

11. (a) If $S = \{1,2,.....10\}$ and a relation R on S where. $R = \{< x, y > | x + y = 10\}$ what are the properties of the relation R?

    (b) If A be a given finite set and $P^{(A)}$ its power set. Let $\subseteq$ be the inclusion relation on the elements of $P(A)$. Draw Hasse diagrams of $< P(A), \subseteq >$ for (i) $A = \{a\}$ (ii) $A = \{a,b\}$, (iii) $A = \{a,b,c\}$, (iv) $A = \{a,b,c,d\}$.

12. (a) If $X = \{1,2,3\}$ and f, g, h and s be functions from X to X given by

$f = \{<1,2>, <2,3>, <3,1>\}$

$g = \{<1,2>, <2,1>, <3,3>\}$

$h = \{<1,1>, <2,2>, <3,1>\}$

$s = \{<1,1>, <2,2>, <3,3>\}$

find $h \circ g$; $g \circ f$; $f \circ h \circ g$;

(b) Show that $f(A \cup B) = f(A) \cup f(B)$.

13. (a) Prove that a subset $S \neq \phi$ of G is a subgroup of $< G*)$ iff for any pair of elements $a, b \in S$, $a * b^{-1} \in S$.

(b) Prove that every finite group of order n is isomorphic to a permulation group of degree n.

14. (a) If g be a homomorphism form a group $< G, * >$ to a a group $< H, \Delta >$, and if K be the Kernel of g and $H' \subseteq$ be the image set of g in H. Then prove that G/K is isomorphic to H'.

(b) If $a, b \in R$ where $< R, +, \bullet >$ is a ring, then prove that $(a+b)^2 = a^2 + ab + b.a + b^2$, where $a^2 = a \cdot a$ and $b^2 = b \cdot b$.

15. Discuss in detail about matrix representation of a graph.

DE-2992

Sub. Code
302

DISTANCE EDUCATION

M.C.A./M.C.A. (Lateral) DEGREE EXAMINATION, DECEMBER 2017.

Third Semester

COMPUTER NETWORKS

(2010 Academic year onwards)

Time : Three hours Maximum : 100 marks

PART A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. Distinguish between guided and unguided transmission media.

2. Describe about the collision free protocol.

3. Explain the virtual circuit switching techniques.

4. Differentiate between the TCP and UDP.

5. Explain in detail symmetric key cryptography.

6. Describe any two flow control mechanisms.

7. Write a short note on Firewall.

8. Explain the Lossy and the Lossless Compression techniques.

PART B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Illustrate the structure of the OSI reference model.

10. Explain 3-bit sliding window protocol with suitable example.

11. (a) Describe the various Internet protocols and its function.

    (b) Write a note on ICMP.

12. Explain the different fields and header in UDP

13. Describe about Electronic mail using SMTP.

14. Discuss the techniques for congestion control in network layer.

15. Explain the following :

    (a) DNS

    (b) WWW.

———————

DE–2993

Sub. Code
303

DISTANCE EDUCATION

M.C.A./M.C.A. (Lateral) DEGREE EXAMINATION, DECEMBER 2017.

Third Semester

SOFTWARE ENGINEERING

(2010 Academic Year Onwards)

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. What are the characteristics of software? Explain.

2. Discuss the software process framework.

3. How will you measure the quality of software? Explain.

4. How to conduct review meetings? Explain.

5. What are the characteristics of good design? Explain.

6. Describe the performance testing.

7. What are SQA activities? Explain.

8. Explain about forward engineering.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Explain the 'Boehm's Spiral Model' with suitable diagram.

10. Discuss about the various parameters associated with design process.

11. Explain the COCOMO – Estimation model in detail.

12. What is prototyping? How prototype models are prepared for a software process? Discuss.

13. Explain in detail about procedural design.

14. Explain software testing strategies.

15. (a) Explain the importance of coupling and cohesion in system design.

    (b) Discuss on white box testing.

DE–2994

Sub. Code
304

DISTANCE EDUCATION

M.C.A./M.C.A. (Lateral) DEGREE EXAMINATION, DECEMBER 2017.

Third Semester

VISUAL PROGRAMMING

(2010 Academic Year onwards)

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. What is an object? Explain its significance.

2. Explain the uses of dialog boxes.

3. What are the various user events in visual programming?

4. Explain the grid control.

5. Explain the data control in VB.

6. Describe the MDI form in VB programming.

7. Write short notes on dynamic data transfer functions.

8. What is OLE? Explain.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Discuss the features of VB-IDE.

10. Explain procedure and function with suitable example.

11. What are the visual C++ components? Explain any three components in detail.

12. What is common dialog box control? Explain all the methods and properties with examples.

13. List and explain the mouse and keyboard events.

14. Discuss about object Linking and Embedding.

15. Explain :

    (a) DLL

    (b) ODBC.

DE–9848

Sub. Code
305

DISTANCE EDUCATION

M.C.A./M.C.A.(Lateral) DEGREE EXAMINATION, DECEMBER 2017.

Third Semester

NETWORK LAB

(2010 Academic Year Onwards)

Time : Three hours Maximum : 100 marks

Examiner has to choose ONE question to each candidate by lot system

Each sub division carries 50 marks.

Break-Up of Marks

Record Note Book	:	5
Algorithm/Flow Chart	:	15
Program	:	35
Debugging & Execution	:	35
Result	:	10
Total	:	100

1. (a) Write a RMI program to add two numbers.

   (b) Write a network program to implement chatting application.

2. (a) Write a Java program to setup a chat server and chat client.

(b) Write a RMI program to find the factorial of given number.

---------- Cut here ----------

3. (a) Write a network program to send the message from one system to another.

(b) Write a java program to encrypt and decrypt a given word.

---------- Cut here ----------

4. (a) Write a network program to display the user name and IP address.

(b) Write a java program to setup time server and time client.

---------- Cut here ----------

5. (a) Write a network program to read a file from the remote system.

(b) Write a java program to setup echo server and echo client.

DE–9849

Sub. Code
306

DISTANCE EDUCATION

M.C.A./M.C.A. (Lateral) DEGREE EXAMINATION, DECEMBER 2017.

Third Semester

VISUAL PROGRAMMING LAB

(2010 Academic Year onwards)

Time : Three hours Maximum : 100 marks

Examiner has to select ONE question and give it to the student by the lot system.

1. (a) Write a VB project to receive a date of birth in a text box and to display the age.

   (b) Write VC++ win 32 Application Program using MFC that receives a new font.

---------- Cut here ----------

2. (a) Develop an application for a calculator performing addition, multiplication, division, subtraction and percentage operations using command buttons.

   (b) Create a window using C++ sources files.

---------- Cut here ----------

3. (a) Write a VB program to create a application to insert, update, delete and retrieve records from a database.

   (b) Write a Visual C++ program to displays current mouse coordinates in status bar.

4. (a) Write a VB project for commercial bank operations using SB account database, with the following features.

   (i) ADD NEW ACCOUNT

   (ii) DEPOSIT ACCOUNT

   (iii) WITHDRAW AMOUNT (with minimum balance condition).

   (b) Write a visual C++ program to create document-view-frame-menu using APP wizard.

---------- Cut here ----------

5. (a) Write a VB project that stores 10 employee records with fields EMPNO, NAME, AGE, SEX and SALARY in an array. Display data fields in text boxes and provide command buttons to move to desired record.

   (b) Write Visual C++ win32 Application program using MFC that creates a list box displays name of the states in India.

---------- Cut here ----------

6. (a) Write a VB project using ActiveX control to create a text box that accepts only numeric value. Provide the following properties for the text box : Back color, fore color and text.

   (b) Write Visual C++ win32 Application program using MFC that creates two push buttons OK and CANCEL on the client area. Buttons should respond 10 user click over them and display appropriate message.

DE–2995

Sub. Code
401

DISTANCE EDUCATION

M.C.A./M.C.A. (Lateral) DEGREE EXAMINATION, DECEMBER 2017.

Fourth Semester

ACCOUNTING AND FINANCIAL MANAGEMENT

(2010 Academic Year Onwards)

Time : Three hours Maximum : 100 marks

PART A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. Prepare journal from the following transactions in the books of Sankar & Co. for the month of May 2013.

May 2013		Rs.
May 1	Started business with a capital of	2,00,000
May 2	Paid into bank	60,000
May 4	Goods purchased	3,00,000
May 6	Purchased goods from Anand on credit	20,000
May 7	Paid to Anand	19,600
May 7	Discount allowed by him	400
May 10	Goods sold	4,00,000
May 13	Cash sales	50,000
May 20	Sold to Naveen	25,000
May 22	Purchased goods from Raju on credit	75,000
May 23	Received from Naveen	24,500
May 23	Allowed him discount	500

May 2013		Rs.
May 24	Withdrew from bank for office use	10,000
May 25	Withdraw from bank for personal use	5,000
May 26	Paid to Raju by cheque	30,000
May 28	Rent paid	10,000
May 29	Interest received from bank	8,000
May 31	Paid salaries	50,000

2. Explain the principles of accounting.

3. The total sales (all credit) of a firm are Rs. 6,40,000. It has a gross profit margin of 15 per cent and a current ratio of 2.5. The firm's current liabilities are Rs. 96,000; inventories Rs. 48,000 and cash Rs. 16,000. (a) Determine the average inventory to be carried by the firm, if an inventory turnover of 5 times is expected; (b) Determine the average collection period if the opening balance of debtors is intended to be of Rs. 80,000. Assume a 360-day year.

4. From the following details prepare a statement showing changes in working capital during year 2011 and 2012.

Liabilities	2011 Rs.	2012 Rs.	Assets	2011 Rs.	2012 Rs.
Share capital	5,00,000	6,00,000	Fixed assets	10,00,000	11,20,000
Reserves	1,50,000	1,80,000	Less : Depreciation	3,70,000	4,60,000
Profit and Loss A/c	40,000	65,000		6,30,000	6,60,000
Debentures	3,00,000	2,50,000	Stock	2,40,000	3,70,000
Creditors for goods	1,70,000	1,60,000	Book debts	2,50,000	2,30,000
Provision for tax	60,000	80,000	Cash in hand	80,000	60,000
			Preliminary expenses	20,000	15,000
Total	12,20,000	13,35,000	Total	12,20,000	13,35,000

5. What do you mean by costing? What are its elements?

6. The following information is available from the annual budget of a company manufacturing only one item.

Budgeted output and sales	:	5000 units
Budgeted selling price per unit	:	Rs. 40

Budgeted cost per unit :

Material	Rs. 15	
Direct labour	Rs. 5	
Variable overhead	Rs. 10	
Fixed cost per unit	Rs. 5	Rs. 35
Budgeted profit per unit		Rs. 5

Calculate the break-even point both in terms of the number of units and sales value.

7. Draw up a material purchase budget from the following information:

Estimated sales of a product are 30,000 units. Two kinds of raw materials A and B are required for manufacturing the product. Each unit of the product requires 3 units of A and 4 units of B. The estimated opening balance in the beginning of the next year : finished goods 5,000 units; A, 6,000 units: B, 10,000 units. The desirable closing balance at the end of the next year : finished product, 8,000 units; A, 10,000 units, B, 12,000 units.

8. Define working capital. Explain its various types.

PART B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. The following trial balance is extracted from the books of a merchant on 31st March 2013.

Particulars	Debit (Rs.)	Credit (Rs.)
Furniture and fittings	640	
Motor vehicles	6,250	
Buildings	7,500	
Capital		12,500
Bad debts	125	
Provision for bad debts		200
Sundry debtors and creditors	3,800	2,500
Stock on April 1, 2012	3,460	
Purchases and sales	5,475	15,450
Bank overdraft		2,850
Sales and purchase returns	200	125
Advertising	450	
Interest	118	
Commission		375
Cash	650	
Taxes and insurance	1,250	
General expenses	782	
Salaries	3,300	
Total	34,000	34,000

The following adjustments are to be made: Stock in hand on 31st March 2013 was Rs. 3,250; depreciate buildings @ 5%, furniture and fittings @ 10% and motor vehicles @ 20%; Rs. 85 is due for interest on bank overdraft; salaries Rs. 300 and taxes Rs. 120 are outstanding; insurance amounting to Rs. 100 is prepaid; one-third of

the commission received is in respect of work to be done next year; write off a further Rs. 100 as bad debts and provision for bad debts is to be made at 5% on sundry debtors; and purchases included purchase of furniture Rs. 200, on January 1, 1989. Prepare a trading and profit and loss account for the year ending 31st March 2013 and a balance sheet as on that date.

10. Given below is the summarized balance sheet and profit and loss account of Bell Limited as on 31.12.2009. You are required to calculate: current ratio, quick ratio, fixed assets ratio, debt equity ratio, proprietary ratio, stock turnover ratio, fixed assets turnover ratio, return on capital employed, debtors turnover ratio, debt collection period, creditors turnover ratio, debt payment period, net profit ratio, and operating ratio.

Profit and Loss Account

	Particulars	Rs.		Particulars	Rs.
To	Opening stock	9,95,000	By	Sales	85,00,000
To	Purchase	54,52,500	By	Closing stock	14,90,000
To	Direct expenses	1,42,500			
To	Gross profit	34,00,000			
	Total	99,90,000		Total	99,90,000
To	Administration expenses	15,00,000	By	Gross profit	34,00,000
To	Selling and distribution expenses	3,00,000	By	Non operating income	90,000
To	Financial expenses	1,50,000			
To	Other non-operating expenses	40,000			
To	Net profit	15,00,000			
	Total	34,90,000		Total	34,90,000

Balance Sheet as on 31.12.2009

Liabilities	Rs.	Assets	Rs.
Issued capital :		Land and	
20,000 shares of		Building	15,00,000
Rs. 100 each	20,00,000	Plant and	
Reserves	9,00,000	Machinery	8,00,000
Creditors	13,00,000	Stock	14,80,000
Profit and Loss		Debtors	7,10,000
account	3,00,000	Cash at bank	3,10,000
6% debentures	3,00,000		
Total	48,00,000	Total	48,00,000

11. What do you mean by budgetary control? Explain the steps involved in budgetary control in detail.

12. From the following data given below, calculate the material price variance, the material usage variance and material mix variance. Consumption per 100 units of product.

Material	Standard	Actual
A	40 units @ Rs. 50 each	50 units @ Rs. 50 each
B	60 units @ Rs. 40 each	60 units @ Rs. 45 each

13. The fixed cost per month in a factory is Rs. 50,000. The contribution per unit is Rs. 50 for product A and Rs. 25 for B. Which of the following product mixes is most yielding?

    (a) 800 A and 100 B

    (b) 1500 A only;

    (c) 3000 B only;

    (d) 1200 A and 400 B.

14. Prepare a flexible budget for overheads on the basis of the following data. Ascertain the overhead rates at 50%, 60% and 70% capacity.

	At 50% capacity (Rs.)
Variable overheads :	
Indirect material	3,000
Indirect labour	9,000
Semi-variable overheads :	
Electricity (40% fixed 60% variable)	15,000
Repairs (80% fixed 20% variable)	1,500
Fixed overheads :	
Depreciation	8,250
Insurance	2,250
Salaries	7,500
Total overheads	46,500
Estimated direct labour hours	93,000

15. Give a detailed note on the various methods of appraising capital expenditure.

DE–2996

Sub. Code
402

DISTANCE EDUCATION

M.C.A./M.C.A. (Lateral) DEGREE EXAMINATION, DECEMBER 2017.

Fourth Semester

INTERNET PROGRAMMING

(2010 Academic Year Onwards)

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. Write short notes on Security and Encryption.
2. Define PERL and explain it with an example.
3. Explain the features of Java Script with an example.
4. Write a program to implement prime numbers using Vbscript.
5. Explain the <table> tag in HTML.
6. Give a brief note on Netscape Navigator.
7. How do you create ActiveX Control? Explain it.
8. Create a simple webpage using images for online shopping.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. List the advantages and disadvantages of CGI.
10. Discuss the basic concepts of Java Language with examples.

11. Explain in detail about Shockwave and Lingo.

12. Write short notes on the following :

    (a) Web Browser

    (b) Web Server.

13. How do you create a Custom Integrated Application with Multiple Protocols?

14. Explain in detail about Images and Hyperlinks in web pages.

15. Discuss on dynamic graphics.

DE–2997

Sub. Code
403

DISTANCE EDUCATION

M.C.A./M.C.A.(Lateral) DEGREE EXAMINATION, DECEMBER 2017.

Fourth Semester

OBJECT ORIENTED ANALYSIS AND DESIGN

(2010 Academic Year Onwards)

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. Define abstraction. List and explain various kinds in abstraction.

2. How can one know if a given class or object is well designed?

3. Describe meta model. What are use cases and how do you write a use case for an application?

4. Describe the two essential elements of a class diagram.

5. Explore the two straight forward elements that can be added to make interaction diagrams more expressive in the presence of certain complicated patterns of interaction.

6. Describe the applications of the object model and the open issues to be addressed in effectively applying the elements of the object model.

7. Explain metaclass relationship among classes. Give the reasons for establishing relationships between two classes.

8. Define and compare state and activity diagram for an application.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Explain the topology of object based and object oriented programming languages. How are OOA, OOD and OOP related?

10. Describe the incremental and iterative nature of classification. Explain about the two processes in the identification of key abstractions.

11. Explain the overview of unified process with its four phases.

12. Explain any five advanced concepts that are necessary to express analysis or design details that are essential to understand the system.

13. Explain the role of UML in object oriented design and building blocks of UML.

14. List and explain the various kinds of relationship between any two objects.

15. Explain the importance of proper classification with an example.

DE–2998

Sub. Code
404

DISTANCE EDUCATION

M.C.A/M.C.A. (Lateral) DEGREE EXAMINATION, DECEMBER 2017.

Fourth Semester

COMPILER DESIGN

(2010 Academic Year onwards)

Time : Three hours Maximum : 100 marks

PART A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. Explain the Structure of a compiler.

2. Explain the different phases of a compiler.

3. What is left recursion? Explain how left recursion is eliminated.

4. Discuss the method of constructing an NFA from a regular expression with suitable example.

5. Show the quadruples and triples for the assignment statement A : = – B*(C + D).

6. Write short notes on contents of a symbol table.

7. What is meant by ambiguous grammar? Give an example.

8. Explain the problems in code generation.

PART B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. What is the importance of code optimization? Explain.

10. (a) Write short notes on compiler writing tools. (7)

    (b) Construct NFA and its transition table for the regular expression (abc)*. (8)

11. Explain in detail about the Top down parsing.

12. Explain how syntax-directed translation scheme is used for generating intermediate code. Give examples.

13. Explain the data structures used for implementing symbol tables.

14. Explain the following :

    (a) Problems in code generation

    (b) Simple code generator.

15. Explain the loop optimization technique with an example.

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DE–9850

Sub. Code
405

DISTANCE EDUCATION

M.C.A./M.C.A. (Lateral) DEGREE EXAMINATION, DECEMBER 2017.

Fourth Semester

INTERNET PROGRAMMING LAB

(2010 Academic Year Onwards)

Time : Three hours

Maximum : 100 marks

Record Note Book : 5
Algorithm/Flow chart : 15
Program : 35
Debugging and Execution : 35
Result : 10
Total : 100

Examiner has to select ONE question and give it to the students by the LOT system.

1. (a) Write a Java script to create a message window with the confirm button.

   (b) Write a Java program using Applet to display different colors and fonts.

---------- Cut here ----------

2. (a) Write a Java script to create the Order Form to select the house articles.

   (b) Write a Java program to create the frames and its controls using Applet.

3. (a) Write a Java script to create a Color palette and display the text in the color chosen from the palette with proper background color.

   (b) Write a Java program to display the dialogue menu using applet.

---------- Cut here ----------

4. (a) Write a VB Script to do the following :

   (i) Check the given password

   (ii) Change the existing password

   (b) Write a Menu-Driven Java program to perform database functions using database connection facility. The functions include display, edit, insert and delete a records from a table.

---------- Cut here ----------

5. (a) Write a VB Script to do the following :

   (i) Display the current system date

   (ii) Find the difference between two dates

   (iii) Find the age of a person by providing date of birth

   [Note : use the date in dd/mm/yy format]

   (b) Write a Java applet to display any 3 images when 3 buttons in the border layout are clicked. The image should be displayed at the center.

DE–9851

Sub. Code
406

DISTANCE EDUCATION

M.C.A./M.C.A. (Lateral) DEGREE EXAMINATION, DECEMBER 2017.

Fourth Semester

COMPILER DESIGN LAB

(2010 Academic Year Onwards)

Time : Three hours Maximum : 100 marks

Examiner should select and give one question to each candidate by lot system.

1. Write a program to convert an infix expression into postfix notation and three address code. Check your program with the following expression $a + b * (c - d)/(e + f)$.

---------- Cut here ----------

2. Write a program to construct canonical collection of LR(0) items for the following grammar

$E \to E + T \mid T, T \to T * F \mid F, F \to (E) \mid id$

---------- Cut here ----------

3. (a) Write a C program to construct a canonical CR parsing table for the grammar

$S \to CC, C \to Cc / d$

(b) Write a C program to recognize constant in the given source code.

4. (a) Write a C program to construct left most derivations to derive word $(a,(a,a))$ using the grammar. $T \to S, S \to a, S \to (T), T \to T, S$.

(b) Write a C program to remove spaces in a given source code.

---------- Cut here ----------

5. (a) Write a C program to check the production rule whether it is an operator grammar or not.

$E \to E + E \mid E * E \mid E \mid id$

(b) Write a C program to remove comment lines in a given source code.

---------- Cut here ----------

6. (a) Write a C program to check the production whether it is a operator grammar or not

$E \to E\ OPEE \to (E)$
$E \to id$
$OP \to + / - / * //$

(b) Write a C program to remove new lines in the given source code.

7. (a) Write a C program to draw the transition diagram for DFA by accepting regular expression

$(1/0)1+10\ (1/0)^*$.

(b) Write a C program to find the TRAILING set for the grammar.

$S \to a,\ S \to (T),\ T \to T, S, T \to S$.

---------- Cut here ----------

8. (a) Write a program to identify constants to a source program.

(b) Write a program to find LEADING and TRAILING sets for the following grammar

$S \to AbB,\ A \to aBb,\ B \to a/Bba$.

DE–2999

Sub. Code
501

DISTANCE EDUCATION

M.C.A./M.C.A. (Lateral) DEGREE EXAMINATION, DECEMBER 2017.

Fifth Semester

DISTRIBUTED COMPUTING

(2010 Academic Year Onwards)

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE of the following questions.

1. Explain briefly about Resource Sharing in Distributed Computing. What is the Process involved in OS boot UP?

2. Explain how financial trading system is suitable for Distributed system.

3. Discuss about Implementation issues involved in Group Communication.

4. Write about various Protection Scheme that support data from unauthorized access.

5. Write a short note on Distributed file System.

6. Write a note on Naming Services.

7. Briefly describe about Process States.

8. With suitable example explain Time stamp Ordering.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR of the following questions.

9. Explain in detail about various types of System Models.

10. Discuss in detail about Internet Protocols.

11. Discuss briefly about Remote Procedure Call (RPC) with Design Issues and Implementation.

12. Explain :

    (a) Cryptographic Algorithm

    (b) Digital Signature

13. Explain in detail about Election Algorithms in Distributed System.

14. Explain in detail about Flat and Nested Distributed Transaction.

15. Define Replication and Explain in detail about Distributed Multimedia Systems.

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DE–3000

Sub. Code
502

DISTANCE EDUCATION

M.C.A./M.C.A. (Lateral) DEGREE EXAMINATION, DECEMBER 2017.

Fifth Semester

WEB TECHNOLOGY

(2010 Academic Year onwards)

Time : Three hours Maximum : 100 marks

PART A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. Write a short notes on HTML Tables.

2. What is XML? Explain Document Type Definition.

3. Explain in detail about Introspection.

4. Explain about Java Beans in API.

5. Write a note on Servlet Application Program Interface.

6. Write short note on JSP Processing.

7. Explain how Sessions can be maintained using Session tracking?

8. Explain database programming using JDBC.

PART B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Explain in detailed about Cascading Style Sheets.

10. What java script? Explain Dynamic HTML with Java script with an example.

11. What is Java Beans? Explain Bean Info Interface with suitable example.

12. Explain in detail about Servlet Interface and it's Life Cycle.

13. Explain in detailed about JSP Application Development.

14. Discuss about Sharing Session and Application data.

15. Explain the following :

    (a) Javax.sql.* package

    (b) Deploying JAVA Beans in a JSP Page.

DE–3001

Sub. Code
503

DISTANCE EDUCATION

M.C.A./M.C.A. (Lateral) DEGREE EXAMINATION, DECEMBER 2017.

Fifth Semester

.NET FRAME WORKS

(2010 Academic year onwards)

Time : Three hours Maximum : 100 marks

PART A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. Discuss about .NET Enterprise Services.

2. Briefly explain .NET Building Block Services.

3. What is Meta Data and explain Reflection Application Program Interface?

4. Explain about Common Type System and MSIL.

5. Explain Attributes Reflection and its types with an example.

6. What are the steps involved to fill a dataset and what is the difference between “Dataset” and “Data Reader”?

7. What are the basic methods of the data adapter? Explain.

8. Discuss Web based Application Development on .NET Programming.

PART B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Explain the .NET Framework Architecture with neat diagram

10. Explain the execution model of the .NET Framework.

11. Explain how memory management is done for reference and value types by CLR.

12. Describe array list and collections .Net Framework.

13. Explain how to build a client and server in .NET Framework.

14. Write note on :

    (a) Type Discovery

    (b) Viewing of Metadata.

15. Explain in detailed about .net Framework Data Provider with suitable example.

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DE–3002

Sub. Code
504

DISTANCE EDUCATION

M.C.A./ M.C.A. (Lateral) DEGREE EXAMINATION, DECEMBER 2017.

Fifth Semester

DATA MINING AND WAREHOUSING

(2010 Academic Year onwards)

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE of the following questions.

1. Briefly describe about the term concept learning.

2. Compare Data Mining Vs Query Tools.

3. Explain Load Manager and its Architecture.

4. Write a Note on Knowledge Discover in Database(KDD).

5. Write a short note on Extract and Load Process.

6. Brief note on DataMarting with its Three Tier.

7. Brief about Performance impact of Security.

8. Write a note on :

   (a) Test Schedule

   (b) Data Backup.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR of the following questions.

9. Explain detail, about the following

   (a) Data mining

   (b) Data Mining needs from data warehouse.

10. Explain the following :

    (a) Association Rules

    (b) Neural Networks

    (c) Genetic Algorithm.

11. Explain in detail about analysis of the data set using traditional query tools and Visualization Techniques.

12. Describe in detail about System Managers and its five types.

13. Explain in detail about Hardware Architecture?

14. Describe in detail about SLA.

15. Write a short note on :

    (a) Assessing performance

    (b) Tuning the data load.

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DE–9852

Sub. Code
505

DISTANCE EDUCATION

M.C.A./M.C.A. (Lateral) DEGREE EXAMINATION, DECEMBER 2017.

Fifth Semester

.NET LAB

(2010 Academic year onwards)

Time : Three hours Maximum : 100 marks

Examiner should select and give ONE question to each candidate by lot system

1. (a) Write a VB.Net program to accept a string and convert from lower case to Uppercase vice versa.

   (b) Write a VB. Net Program to display the Student mark list using the windows application.

---------- Cut here ----------

2. (a) Develop a VB.Net application using Datagrid to display records.

   (b) Create a currency converter applications in ASP.NET with two input text boxes named UK dollar and Indian rupees and a submit button named convert.

3. (a) Write a VB .Net Program to bind data using Hyperlink column in data grid.

   (b) Create an application that displays the information of all students without selection, edition and sorting options.

---------- Cut here ----------

4. (a) Design a form to select image from list and display it in the picture box

   (b) Develop an Employee payroll system to generate a crystal report from an existing database using VB.NET.

---------- Cut here ----------

5. (a) Write a program to simple calculator using windows application.

   (b) Write an ASP.NET program that gets user input such as the user name, mode of payment, appropriate credit card. After the user enters the appropriate values the validation button must validates the values entered.

---------- Cut here ----------

6. (a) Develop a database application to General store the details of students using ADO.NET

   (b) Write a Program in ASP that has a form taking the users name as input. Store this name in a permanent cookie & whenever the page is opened again, then value of the name field should be attached with the cookies content.

7. (a) Develop a web application for Online Voter Registration.

(b) Write a Program to display the difference between the two dates in the calendar.

DE–9853

Sub. Code
506

DISTANCE EDUCATION

M.C.A./M.C.A. (Lateral) DEGREE EXAMINATION, DECEMBER 2017.

Fifth Semester

WEB TECHNOLOGY LAB

(2010 Academic year onwards)

Time : Three hours Maximum : 100 marks

Examiner should select and give ONE question to each candidate by lot system.

1. (a) To create an html file that link to different html page which contains images, tables, and also linked within a page.

   (b) Create a web page that describes an unordered list of at least five books The bullet for each book must be a small image hot spot. When you click the picture all the related Information about the book must be shown.

---------- Cut here ----------

2. (a) Write an XML document to display Student profile.

   (b) Create a web page for MCA Department Home page. Align the text with colors. Use the links with various pages.

---------- Cut here ----------

3. (a) Write a JSP program to prepare Staff pay bill using JDBC.

   (b) Write an ASP program to prepare stock details in super market using java script.

4. (a) Write a servlet program to display the request and response header information.

   (b) Create an application form for MCA Course using various HTML controls.

---------- Cut here ----------

5. (a) Create a HTML frame using various controls.

   (b) Write a servlet program to create a chatting application.

---------- Cut here ----------

6. (a) Write an XML document to display your favourite Indian cricket players. Write an XSL style sheet and attach this to the XML document to validate the document using DTD or XSD.

   (b) Create an HTML table for your Class time table using Row span and Col span.

---------- Cut here ----------

7. (a) Write an XML document to display Student resume.

   (b) Write a HTML program to pick the color from color palette and display color name in text box.

---------- Cut here ----------

8. (a) To create a servlet program to retrieve the values entered in the html file.

   (b) Write Servlet program to send Email message.

DE–3003

Sub. Code
601

DISTANCE EDUCATION

M.C.A./M.C.A. (Lateral) DEGREE EXAMINATION, DECEMBER 2017.

Sixth Semester

MOBILE COMMUNICATIONS

(2010 Academic Year onwards)

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE of the following questions.

1. Explain the concept of SDMA and FDMA.

2. List out two basics reasons for Handover and also four Possible Handover Scenarios in GSM.

3. Discuss about Infrared and Radio Transmission.

4. Define Bluetooth and explain the User Scenarios for Wireless Pico nets or WPANs.

5. Write a short note on Minimal Encapsulation.

6. Compare IPV4 Vs IPV6.

7. Write a short note on Mobile TCP.

8. Discuss about Session Protocol.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR of the following questions.

9. Explain in detail about GSM System Architecture with neat diagram.

10. Explain about HIPERLAN.

11. Discuss about Ad hoc Network Routing (a) DSDV (b) DSR.

12. Discuss about (a) Fast Retransmission (b) Fast Recovery.

13. Explain with suitable diagram about the Transaction oriented TCP.

14. Discuss about Wireless Telephony Application.

15. Explain in detail about Transport Layer Security.

DE–3004

Sub. Code
602

DISTANCE EDUCATION

M.C.A./M.C.A. (Lateral) DEGREE EXAMINATION,
DECEMBER 2017.

Sixth Semester

MIDDLEWARE TECHNOLOGY

(2010 Academic year onwards)

Time : Three hours Maximum : 100 marks

PART A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. Discuss the 2-tier model of client/server architecture.

2. Explain the 3-tier model of client/server architecture.

3. Briefly discuss the JSP as presentation service.

4. Describe the RMI as interaction service.

5. Explain entity beans with suitable example.

6. Write short notes on message driven beans.

7. Mention the HTML controls.

8. Develop a simple web service.

PART B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Explain the J2EE architecture in detail with neat diagram.

10. With a neat diagram, discuss the DOT NET architecture.

11. Give an overview of data management services.

12. Discuss in detail the EJB component model.

13. Write detailed notes on web server controls.

14. How to develop ASP. NET clients for web services? Explain.

15. Describe any three web services.

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DE–3005

Sub. Code
603

DISTANCE EDUCATION

M.C.A./M.C.A. (Lateral) DEGREE EXAMINATION, DECEMBER 2017.

Sixth Semester

AGENT BASED INTELLIGENT SYSTEM

(2010 Academic Year onwards)

Time : Three hours Maximum : 100 marks

PART A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. List out the various searching strategy.

2. Write a brief note on alpha beta pruning that is used in gaming.

3. Discuss forward and backward chaining.

4. Bring out the significance of logical agents.

5. Describe the partial order planning approach with an example.

6. Write a brief note on multiagent planning.

7. Explain how vagueness is represented using fuzzy logic.

8. Discuss the reinforcement learning.

PART B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Explain about the different Heuristic Search techniques.

10. Discuss the different issues in knowledge representation.

11. Explain the syntax and semantics used in first order logic.

12. Explain how planning is carried out using state space research.

13. Explain the semantics of Bayesian networks.

14. With suitable examples Explain complex decision making.

15. Explain the different statistical learning methods.

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DE–3006

Sub. Code
604

DISTANCE EDUCATION

M.C.A./M.C.A (Lateral) DEGREE EXAMINATION, DECEMBER 2017.

Sixth Semester

OPEN SOURCE ARCHITECTURE

(2010 Academic Year Onwards)

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE of the following questions.

1. Discuss about Common System Configuration files and Log Files.

2. What is the Process involved in OS boot UP?

3. How display devices are detected? Explain.

4. Discuss about GNU Compiler TOOLS.

5. How CVS Supports and Manage Source Code Revisions? Explain.

6. How Graphical Debuggers differs from Memory Debugging? Explain.

7. Write short notes on Dynamically linked libraries.

8. Describe about the concept python Programming.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR of the following questions.

9. Describe the basics of TCP/IP Networking and Routing.

10. Discuss in detail about X Windows system configuration and Utilities and configuration X Windows.

11. Explain :

    (a) Setting up email servers

    (b) Setting up Web Servers.

12. Discuss about the auto conf and auto gen in detail.

13. Explain about Linking against object archives and dynamic shared object libraries.

14. Explain about BASH, SED and AWK Scripting.

15. Explain in detail about X Windows Server Architecture.

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DE–9854

Sub. Code
605

DISTANCE EDUCATION

M.C.A./M.C.A. (Lateral) (Semester) DEGREE EXAMINATION, DECEMBER 2017.

Sixth Semester

OPEN SOURCE PROGRAMMING LAB

M.C.A. (S) (2010 Academic Year Onwards) / M.C.A. (S) (Lateral)

Time : Three hours Maximum : 100 marks

Examiner should select and give ONE question to each candidate by lot system.

1. (a) Write a PHP Programming to create web page for online Cinema Ticket booking.

   (b) Write a PHP Program to insert and modify the data from student's marks table.

---------- Cut here ----------

2. (a) Write a PHP Program to Update the Customer table Using MYSQL.

   (b) Write a PHP Program to create web page for online mobile shopping.

---------- Cut here ----------

3. (a) Write a PHP Program to select top three candidates from the student list using MYSQL.

   (b) Write a PHP Program to insert/ update recent cricket score of the current player.

4. (a) Write a PHP Program for online book shopping using MYSQL.

(b) Write a simple PHP Program to copy the content of one array to another array.

---------- Cut here ----------

5. (a) Write a simple PHP Program to create a cookie and to retrieve the cookie from your computer.

(b) Write a PHP Program for the following

(i) Replace a string

(ii) Concatenation of two strings

(iii) Reverse a string.

---------- Cut here ----------

6. (a) Write a Program to create simple login form, submit the user details such as username and password using POST method and authenticate the user from server side using PHP.

(b) Write a PHP Program for the following

(i) Copy a string

(ii) Compare a string.

7. (a) Write a Program to design a web page for your dream company using PHP and HTML.

   (b) Write a PHP Program to sort the content of an array in descending order.

---------- Cut here ----------

8. (a) Write a Program to create a simple credit card payment details using PHP and HTML.

   (b) Write a PHP Program to upload a file to the server.

DE–9855

Sub. Code
606

DISTANCE EDUCATION

M.C.A./M.C.A. (Lateral) DEGREE EXAMINATION, DECEMBER 2017.

Sixth Semester

MULTIMEDIA TOOLS LAB

(2010 Academic Year Onwards)

Time : Three hours Maximum : 100 marks

Break-up marks

Record : 5

Algorithm/Flow chart : 15

Program : 35

Execution : 35

Result : 10

Total : 100

Examiner has to choose any ONE question for each candidate by lot system.

1. (a) Develop an animation in Flash to implement Masking.

   (b) Develop an animation using FLASH to create a rainfall effect.

2. (a) Implement a program in FLASH to develop a bouncing ball Animation.

   (b) Implement an animation to show a flying bird in the sky.

---------- Cut here ----------

3. (a) Implement a program in flash forgetting rainfall effect.

   (b) Implement a program to create a Maze using Maya.

---------- Cut here ----------

4. (a) Develop an animation to show a growing flower plant.

   (b) Implement a program for animating flight with twin tower using flash.

---------- Cut here ----------

5. (a) Create a cartoon with Laughing face using FLASH.

   (b) Animate your name with different colours. Make the characters to blink.

---------- Cut here ----------

6. (a) Implement a program in FLASH for bouncing a ball with a pad moving along with the ball.

   (b) Develop an animation in FLASH to glow a bulb with changing colours.

7. (a) Create any two different shapes animate the shapes slowly appear and disappear one after another.

   (b) Develop a animated banner for web page using dream viewer for a car demonstration.

---------- Cut here ----------

8. (a) Animate your name to create the Rotating effect in clock wise directions and display the text in reverse format.

   (b) Animate your own text to create the horizontal flipping effect.

---------- Cut here ----------

9. (a) Develop an animation to show a flying butterfly in a garden.

   (b) Develop an animation in flash to show a moving flight in the runway.

---------- Cut here ----------

10. (a) Implement a program to animate the flight in a runway.

    (b) Develop a flash animation for motivating eye donation.